Astrophotography

for beginners

And the rest...

Course plan

Week	First half of evening	Break	Second half of evening
1	Introduction & basics	*Cafe is open 6pm-8:30pm!*	Basics inside PHP
2	Wide angle photography		Targets in the planetarium
3	Processing Software		Sun, Moon & planets
4	Zoom lenses & telescopes		Video astronomy (time permitting)
5	Small 'scope observing		Observing with 28"
6	And finally...		Participants photo's

Tracking mounts

Vixen Polarie ~£400

Tracking mounts

AstroTrac ~£450

Tracking mounts

Losmandy StarLapse ~ 800 euro

Tracking mounts

Orion Mini EQ1 ~£150

Orion

Orion

Camera additions

Telescope adapters for DSLR cameras

• A range of eyepieces can be bought for telescopes with different focal lengths and so different magnifications. They are usually 1¼” in diameter

DSLR telescope adapters come in two parts:

(1) camera-to-T-mount adapter

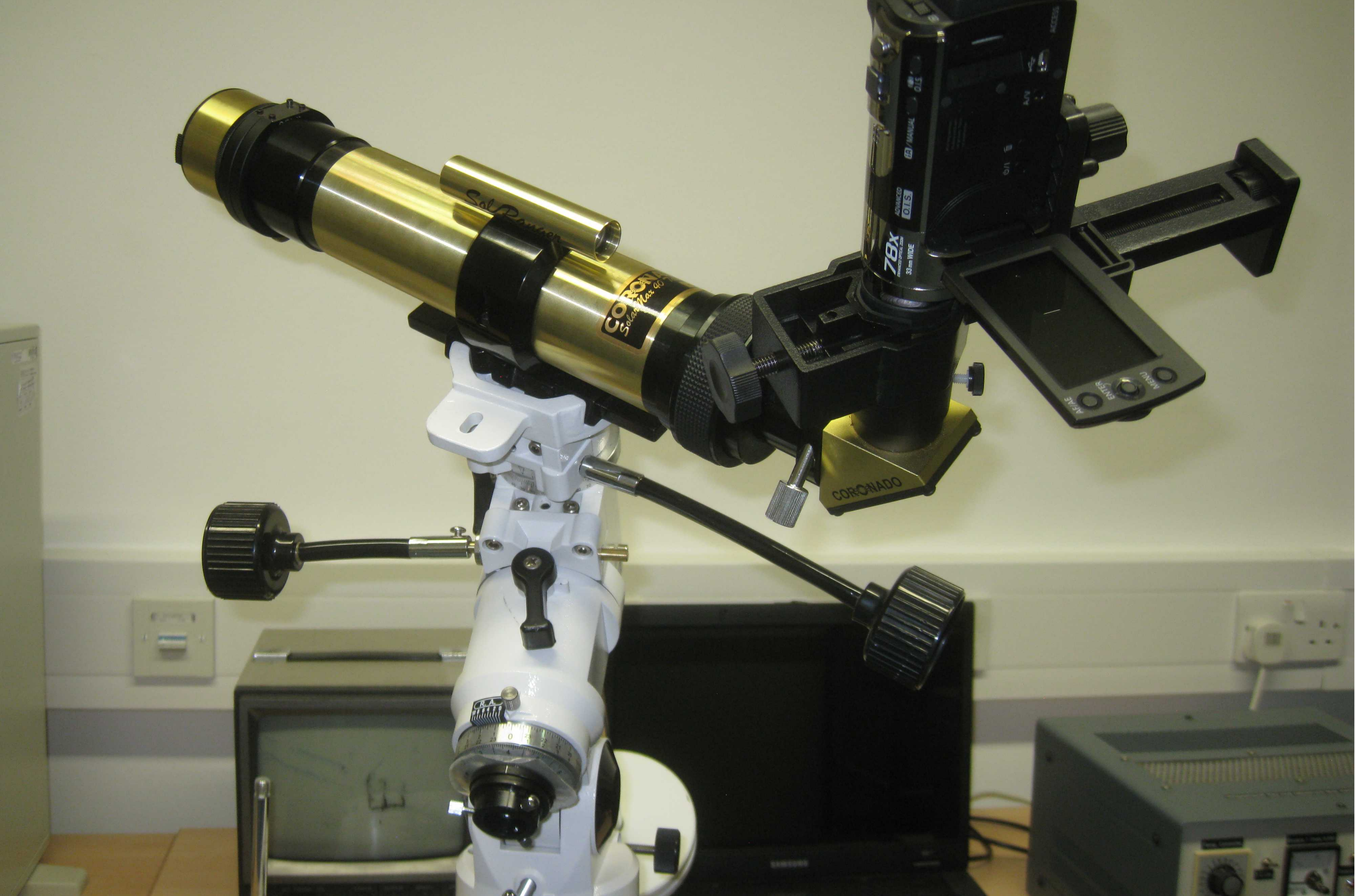
(2) T-mount-to-telescope 1¼” adapter

For examples & prices, see e.g.

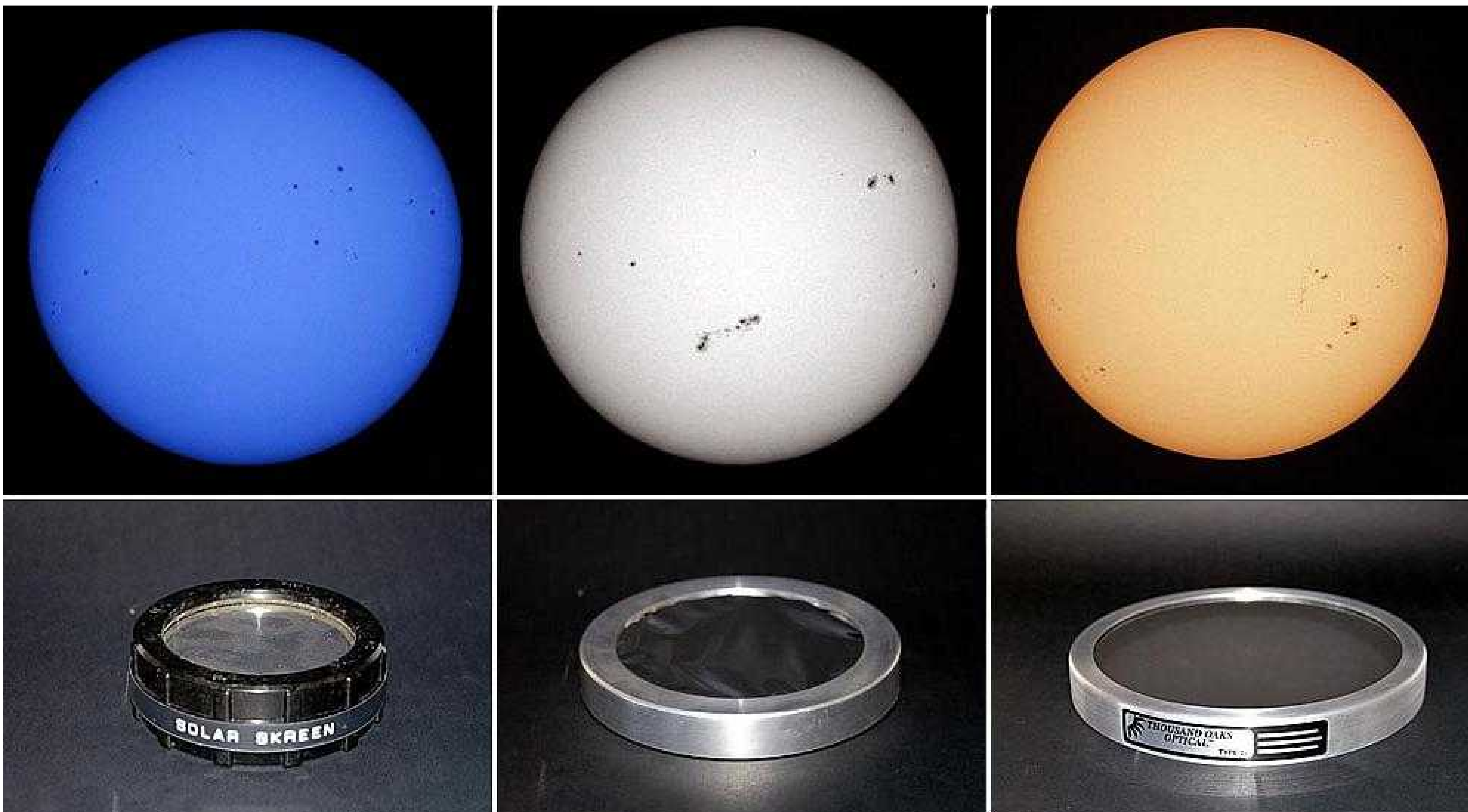
www.harrisontelescopes.co.uk/acatalog/DSLR_Camera_Adaptors.html

Video cameras

Video cameras



Filters



Polarising filters
- reduce the glare and scatter of the sky

Can be used for observing the Moon

Photographic & Astronomical Filters

Camera additions

Light pollution filters

Steve Richards,The Chanctonbury Observatory (Hutech IDAS LP filter)

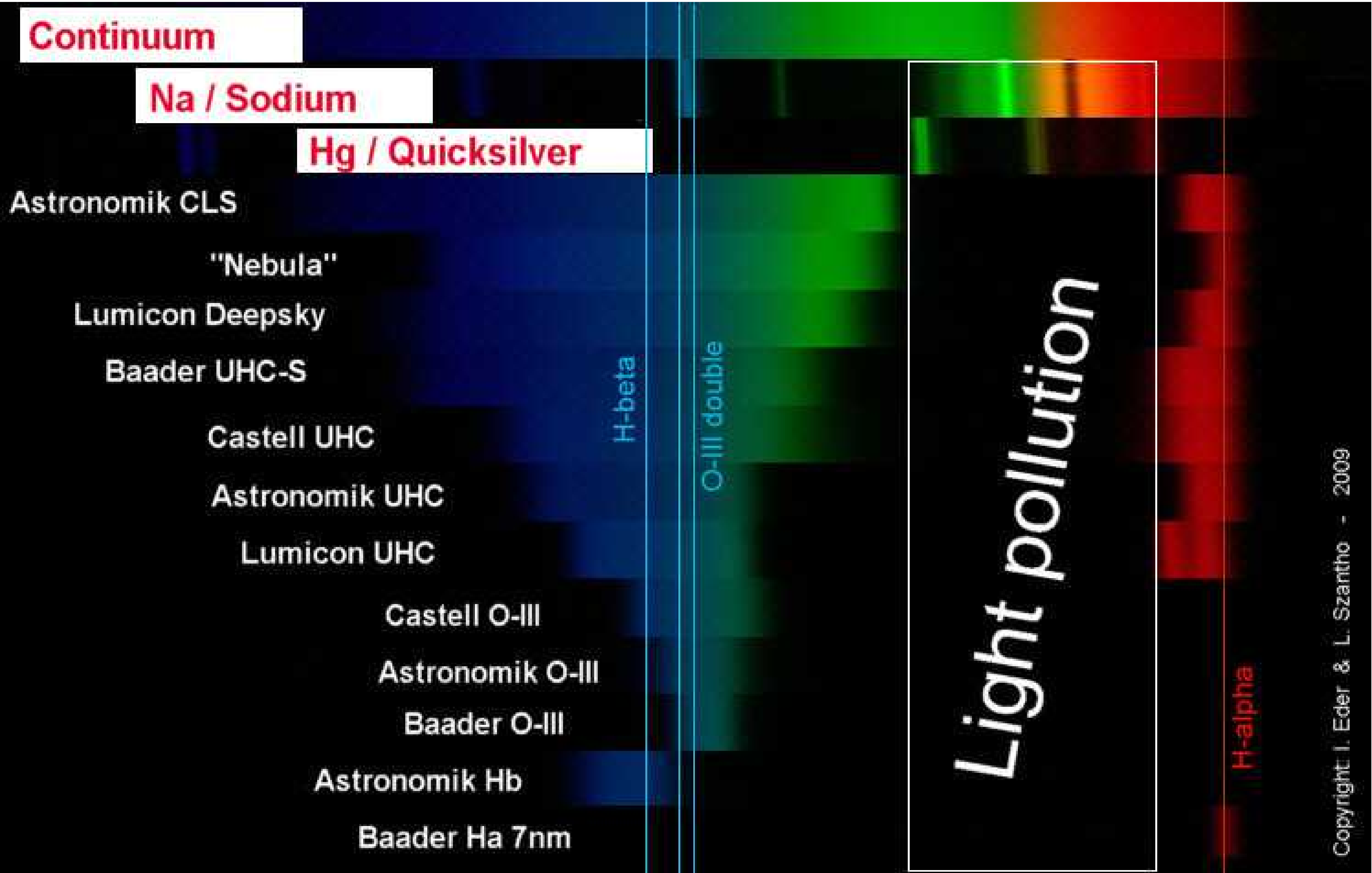
Right:
Brian Thompson

Below:
Koen van Gorp

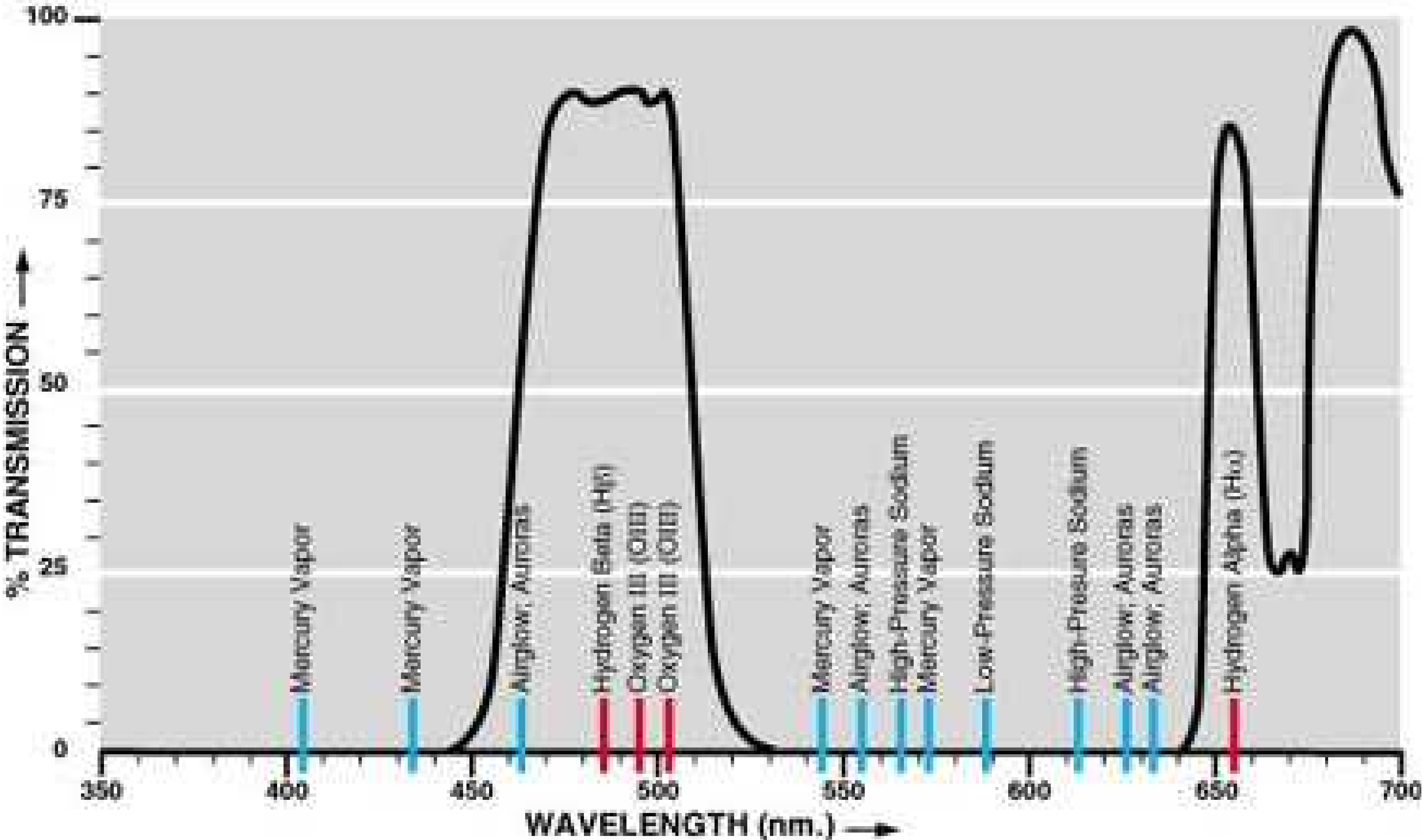
Hutech Astronomical Products

Nebula filters

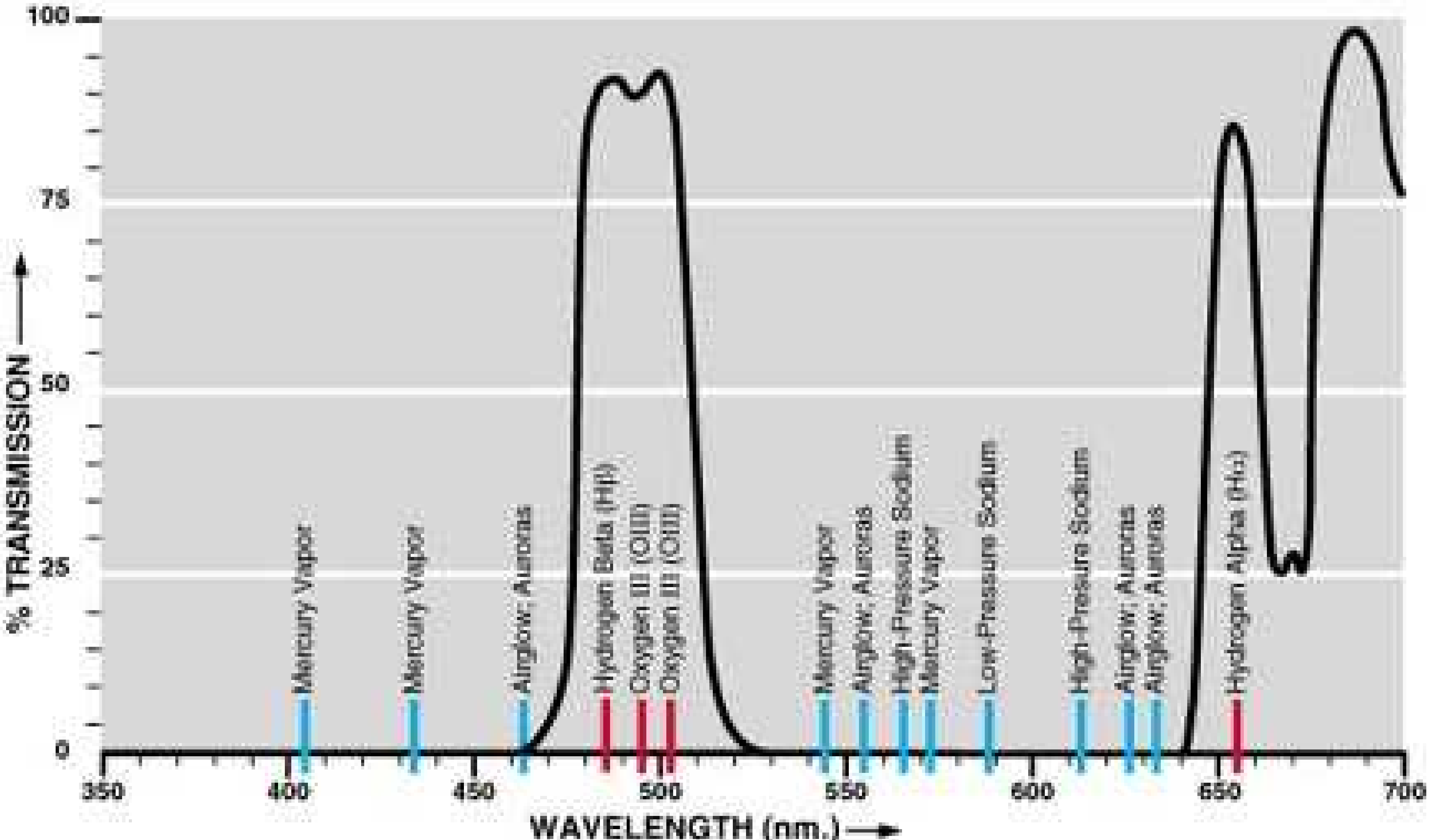
Spectral properties of astronomical filters



Broadband filters
Meade 4000 broadband

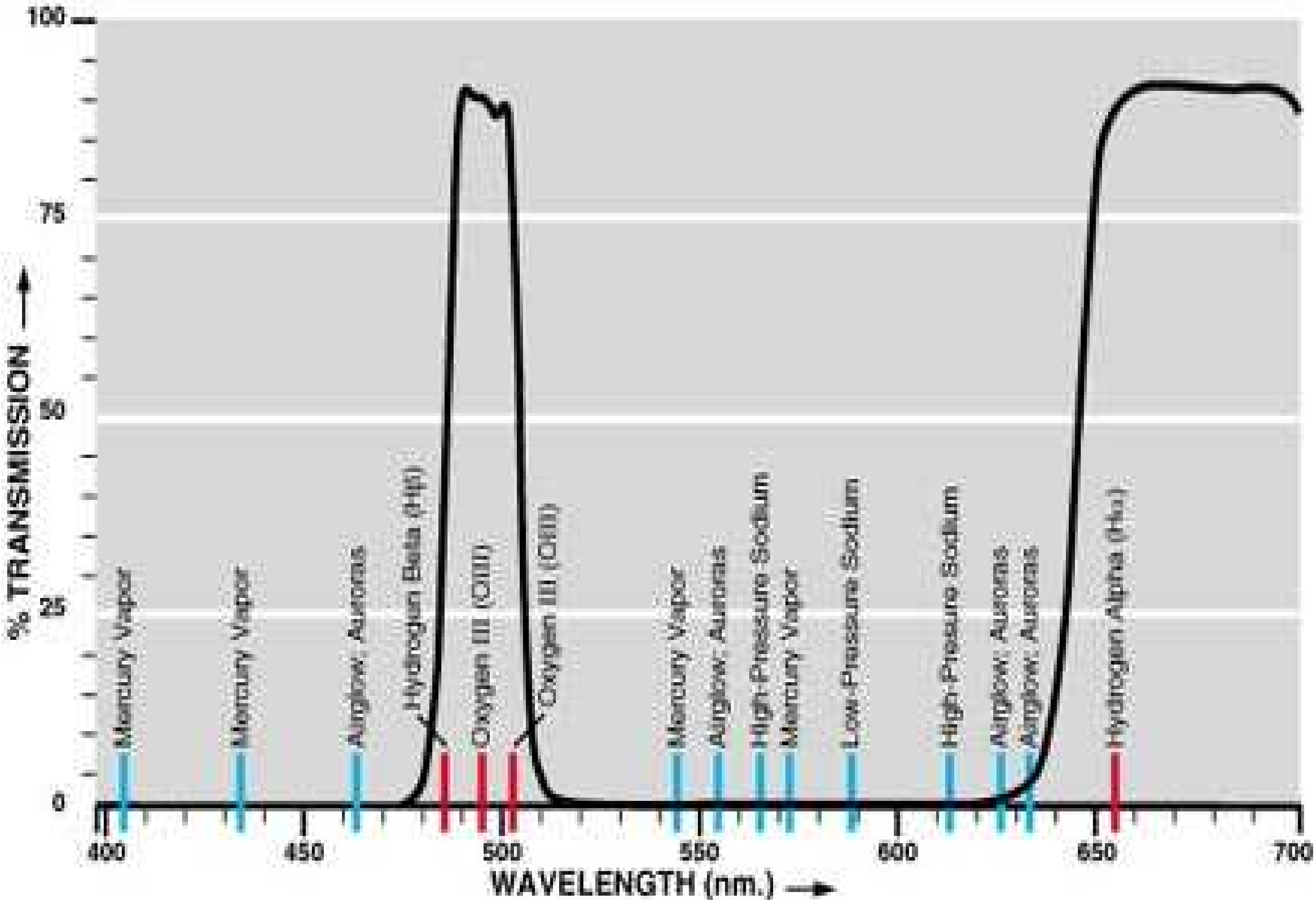


Narrowband filters
Meade 4000 narrowband nebula filter



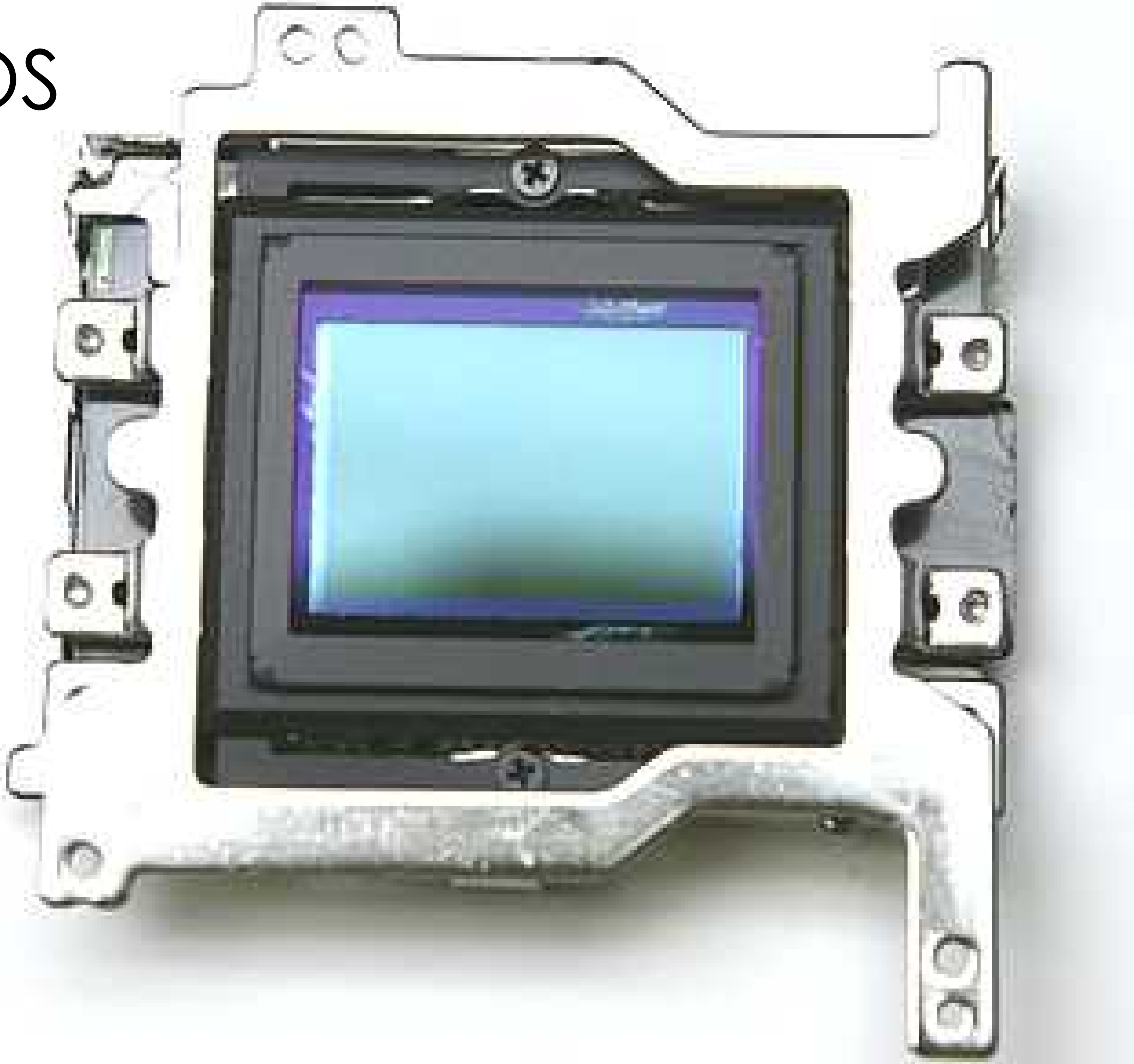
Narrowband filters

Meade 4000 narrowband O-III nebula filter

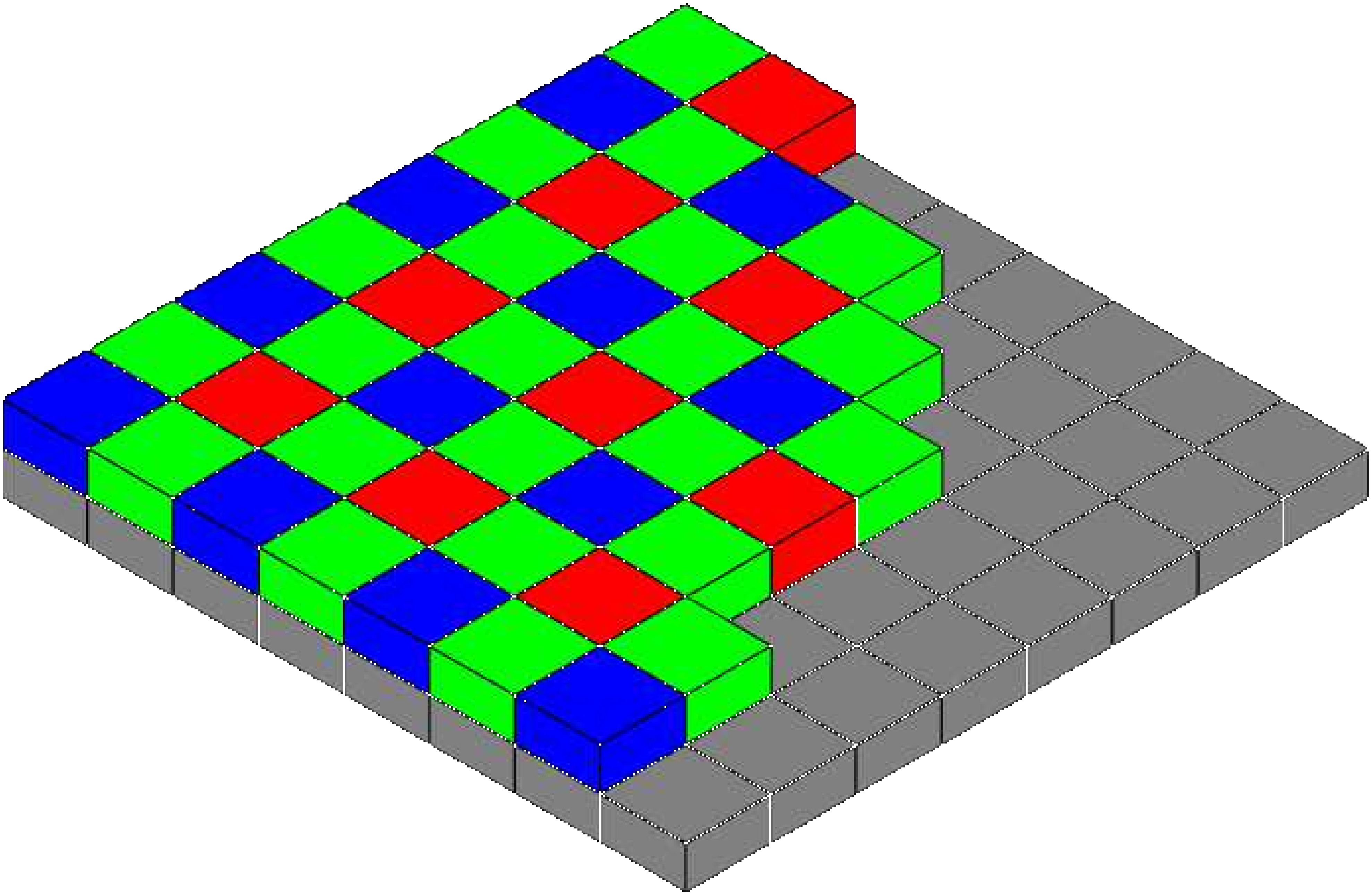


CMOS Vs CCDs

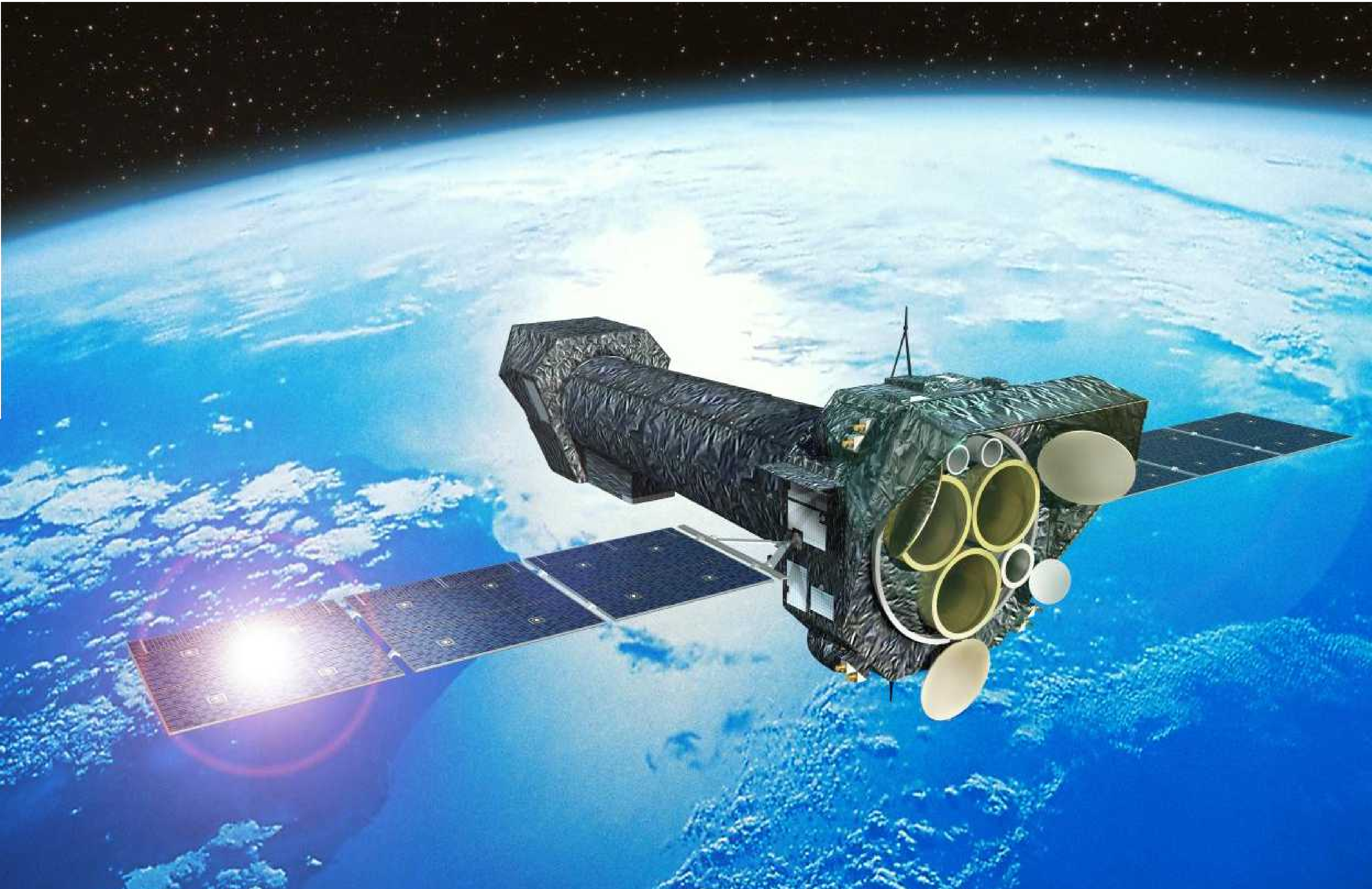
CMOS



Colour sensitivity



XMM-Newton



CMOS v CCD's

CCD cameras

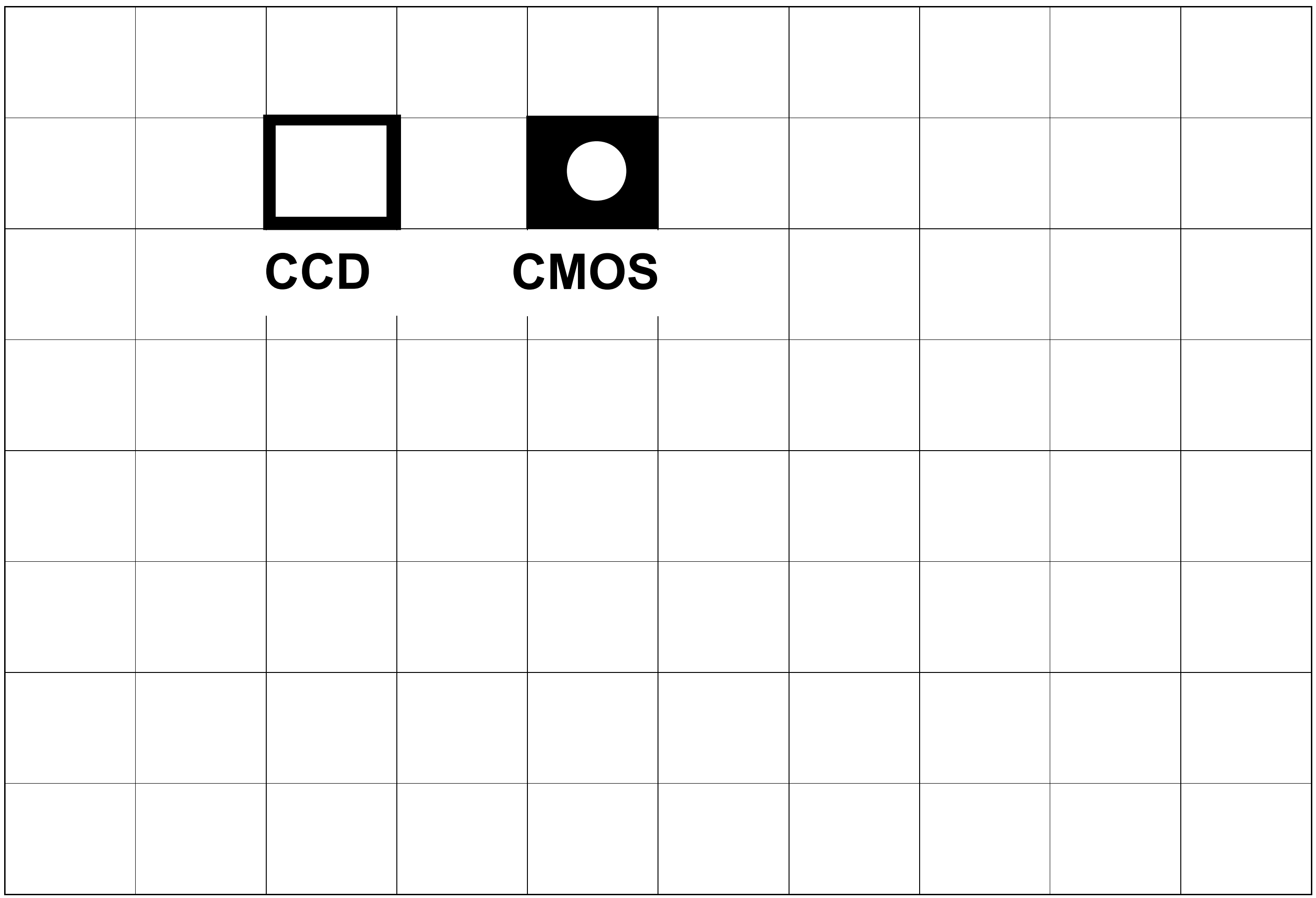
CCD chips have less noise than a CMOS

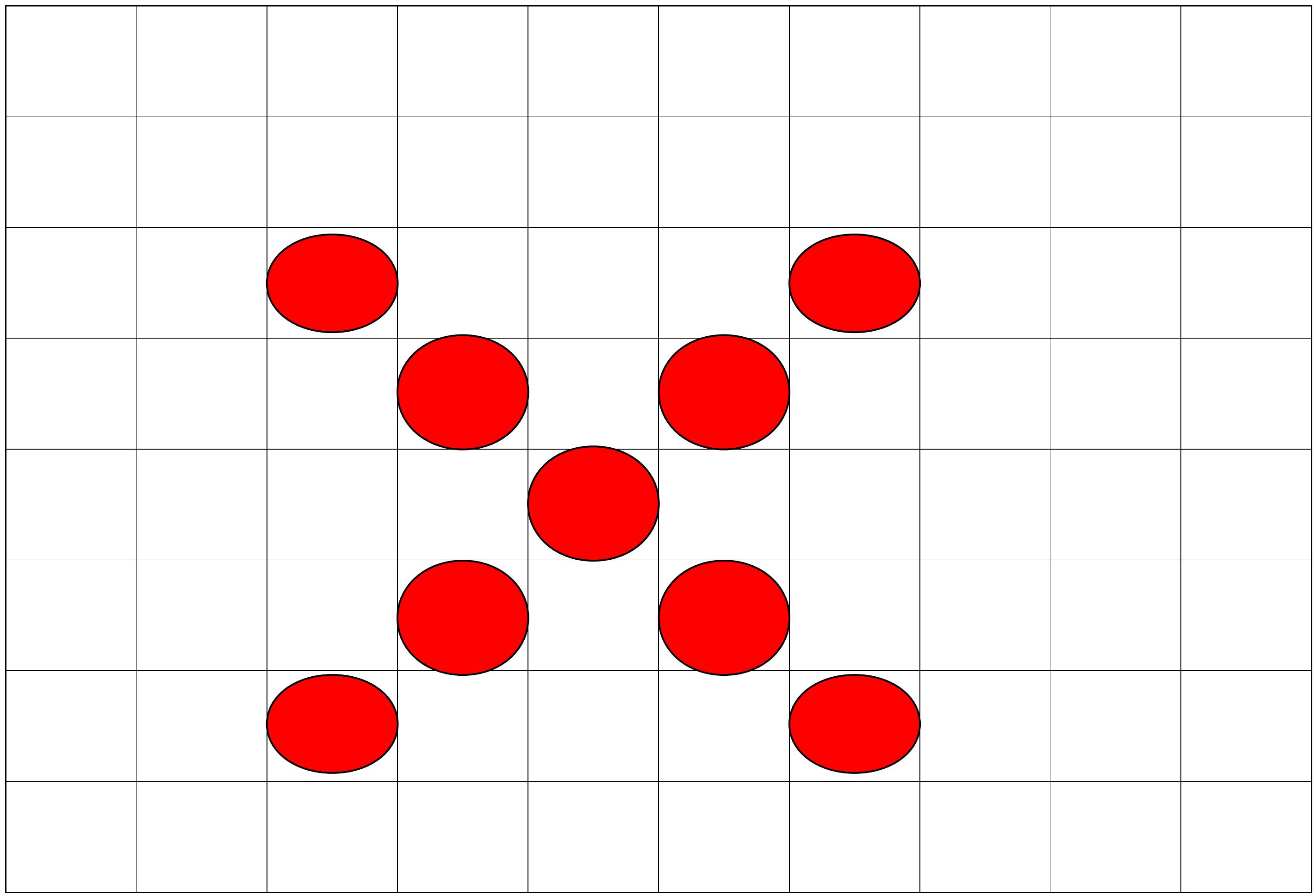
CCDs are more sensitive (as electronics on each pixel)

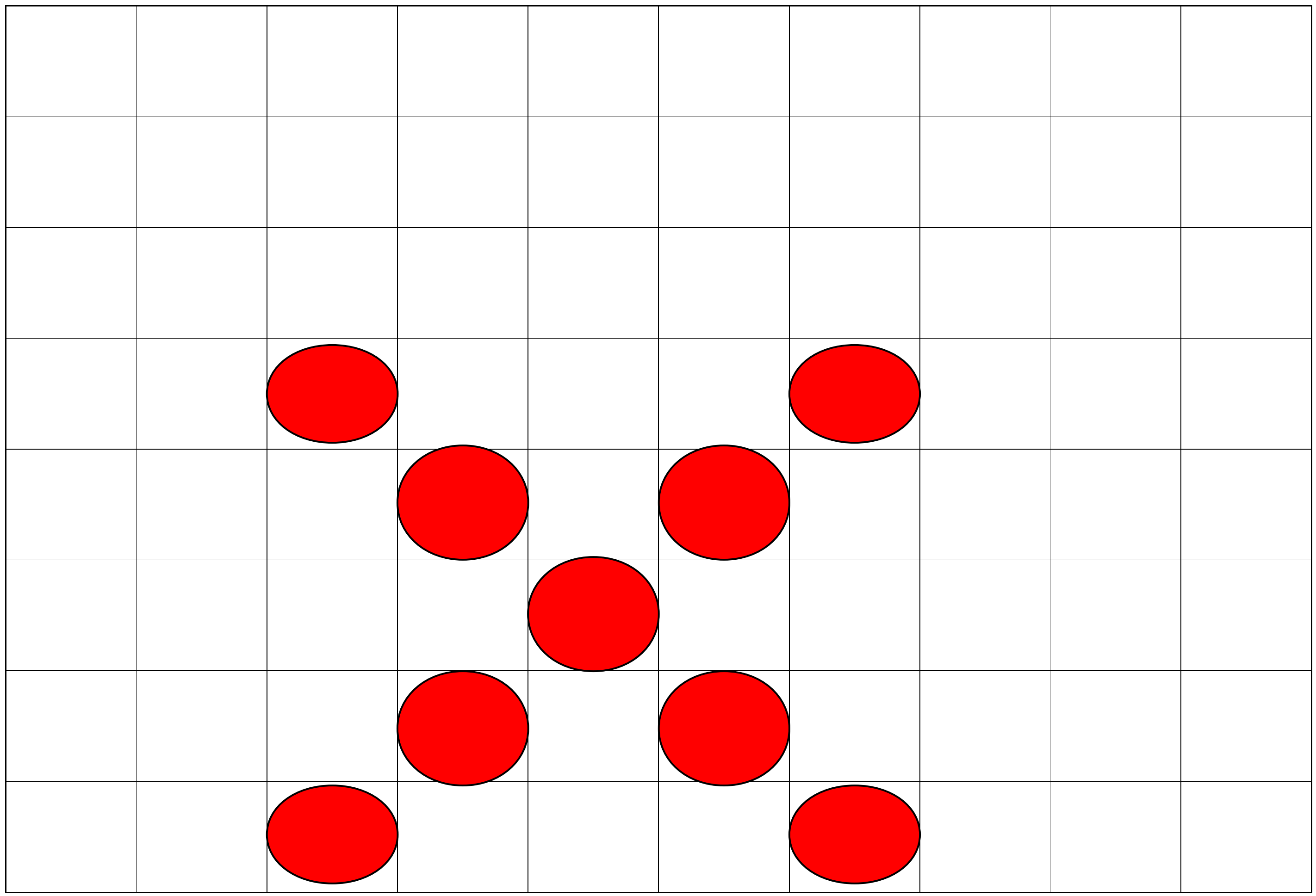
And so are usually more expensive...

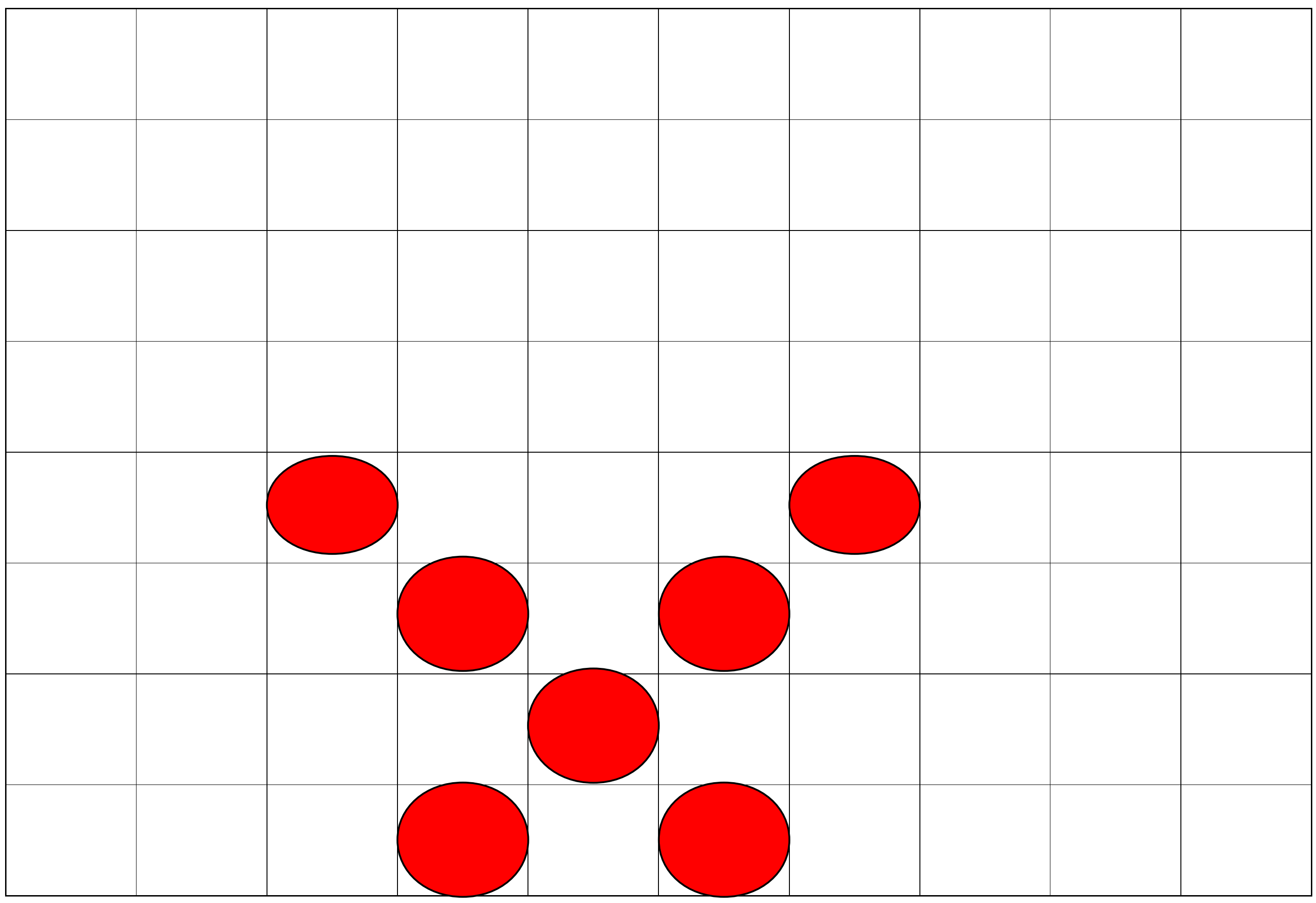
CMOS: Charge is read directly out. Cheap, but imprecise & lack sensitivity.

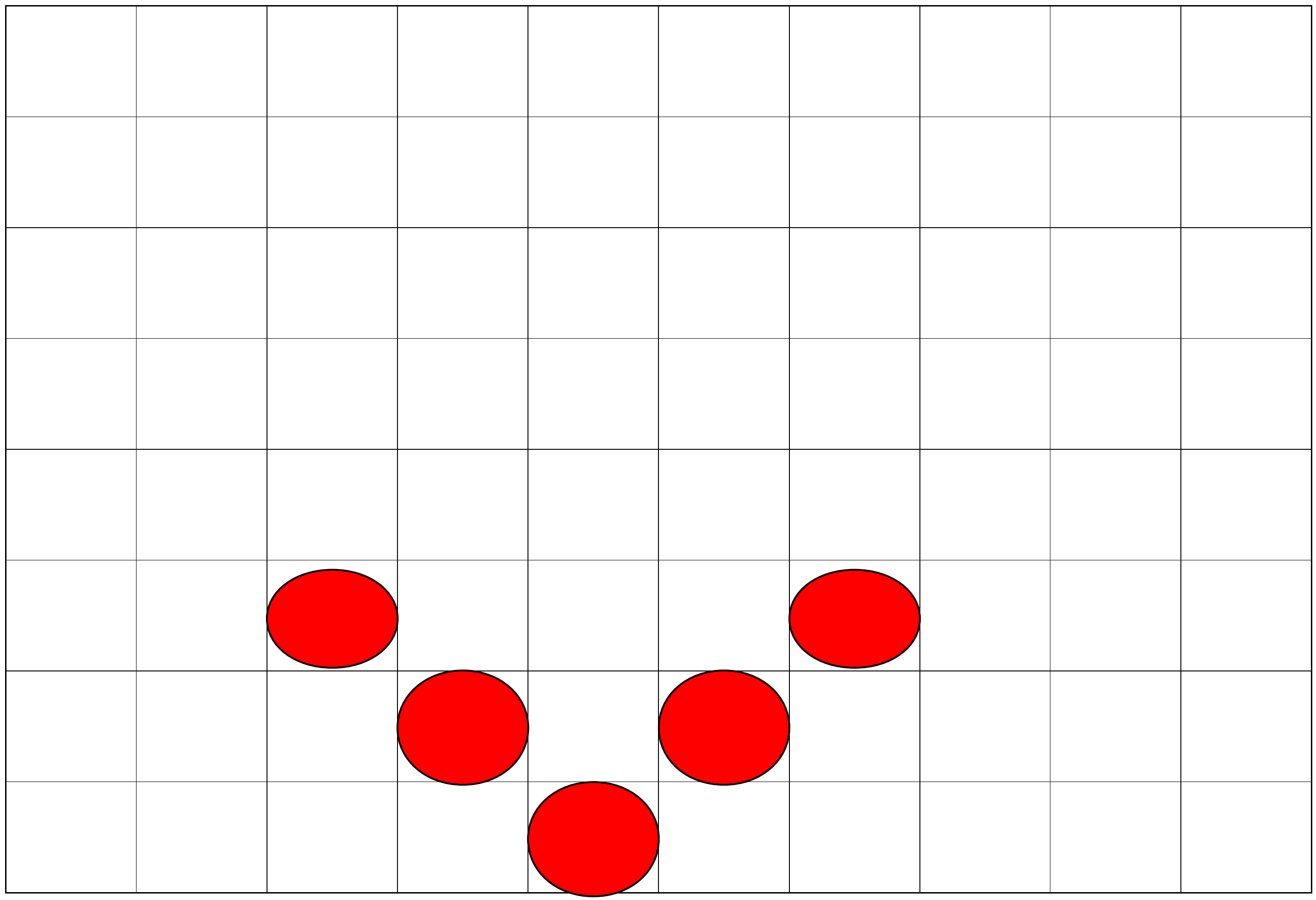
- CCD: Charge is read out of one point, line by line. Expensive as each pixel must be close to identical in manufacture, but sensitive & minimal noise.

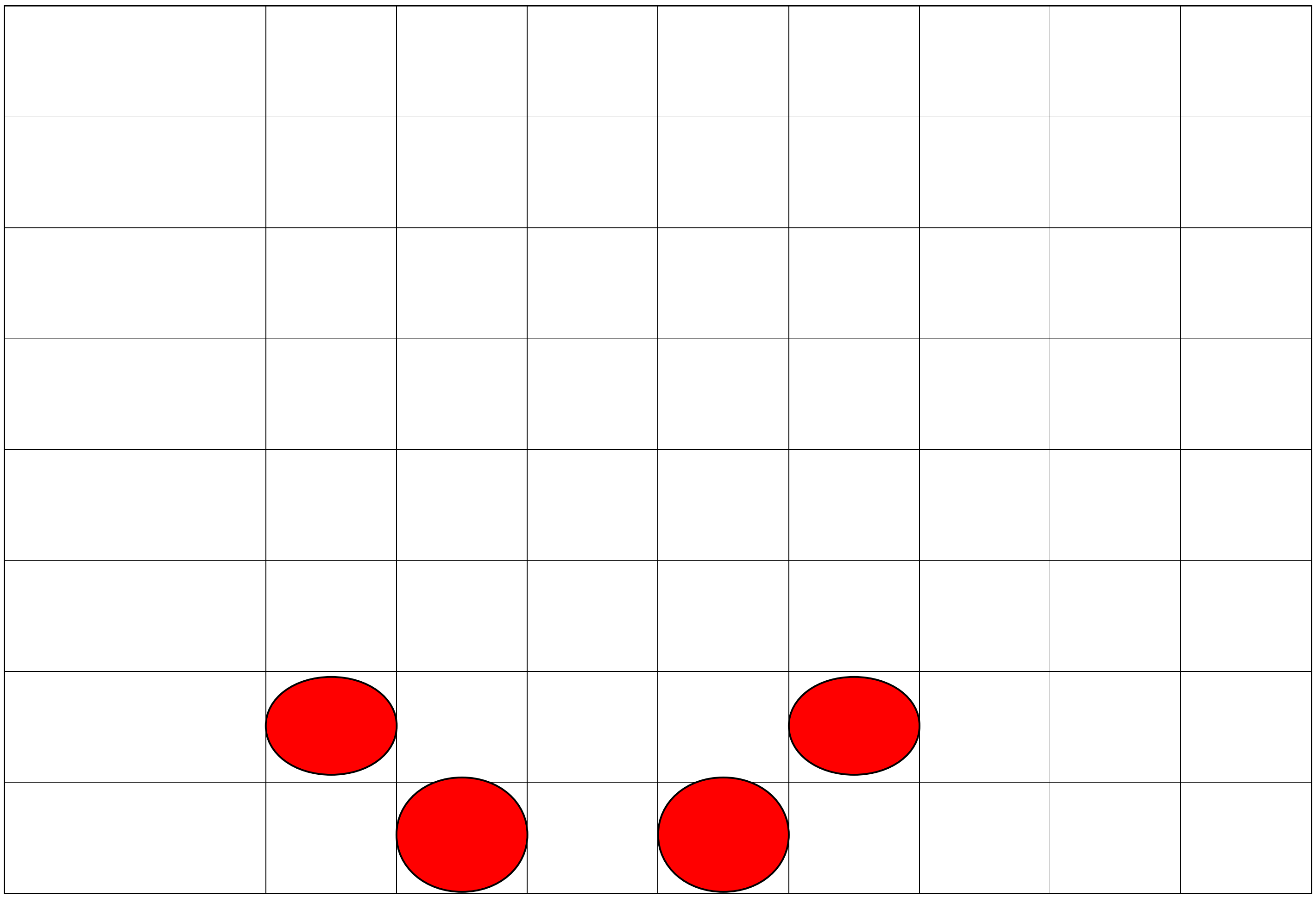


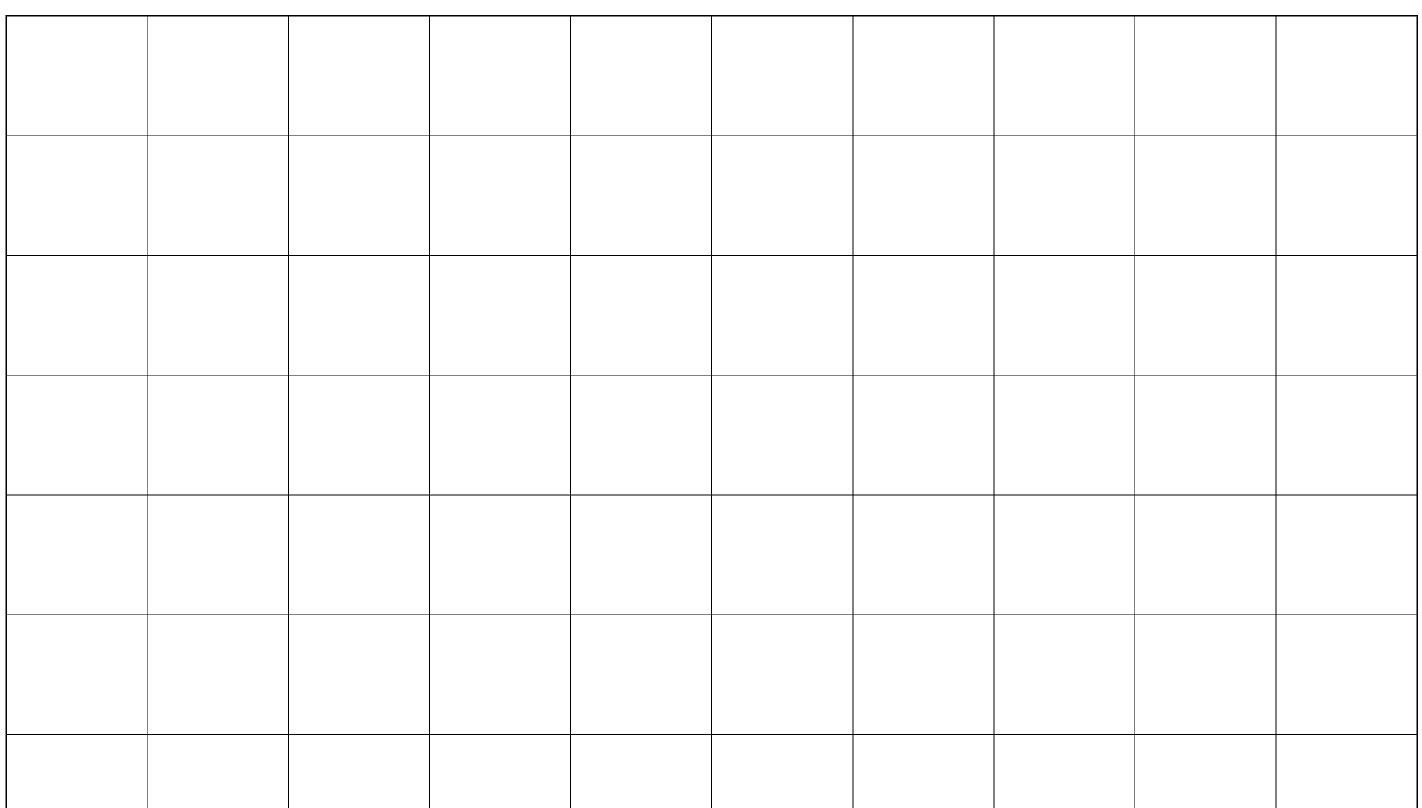






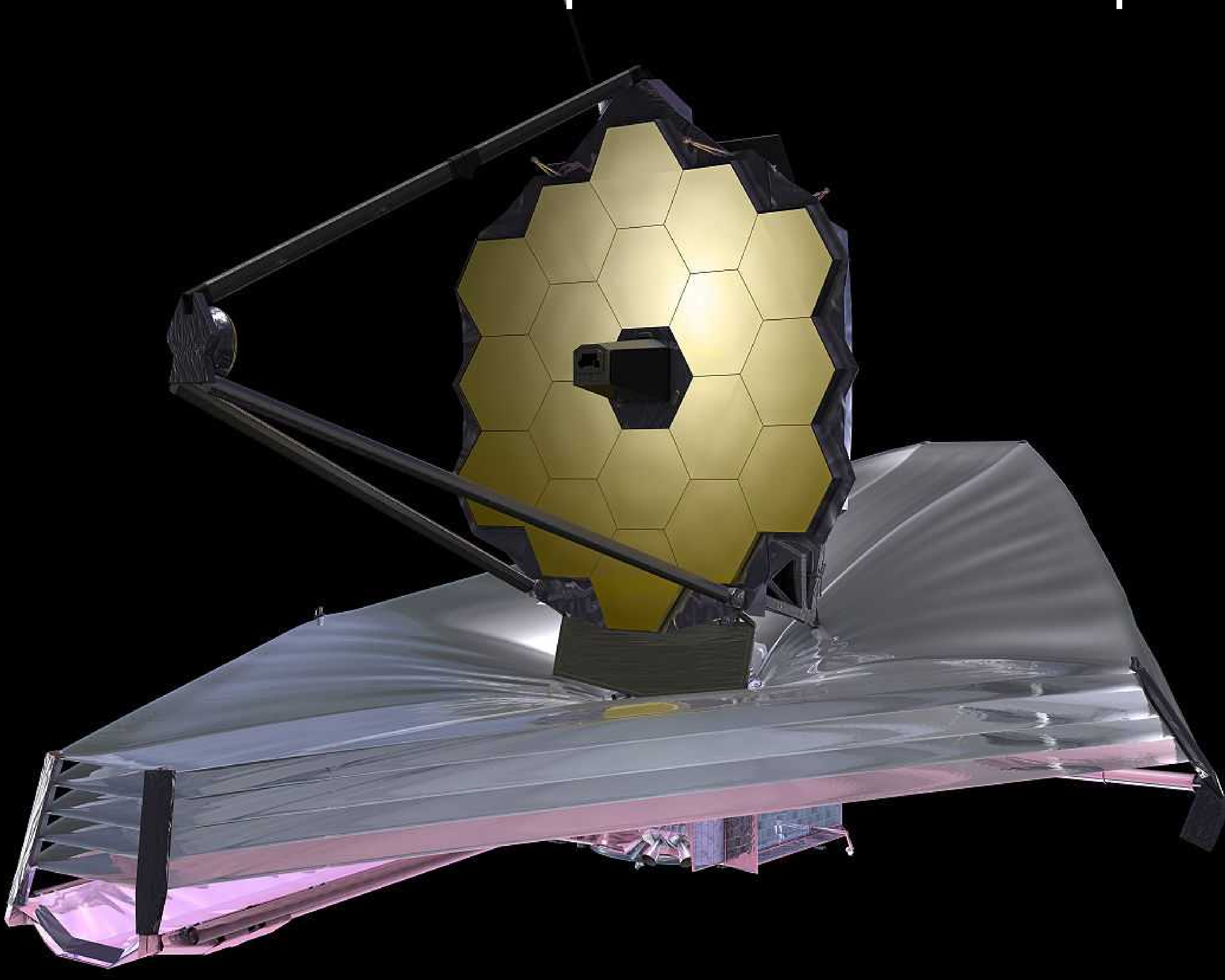






MIRI filter wheel

James Webb Space Telescope



Useful Apps

02-UK 11:00
5 Day Forecast
Newcastle
6°
5°
4°
5°
Weather

02-UK 11:01
Met Office
Google

BBC WEATHER
Nottingham
08 00
09 00
10 00
11 00
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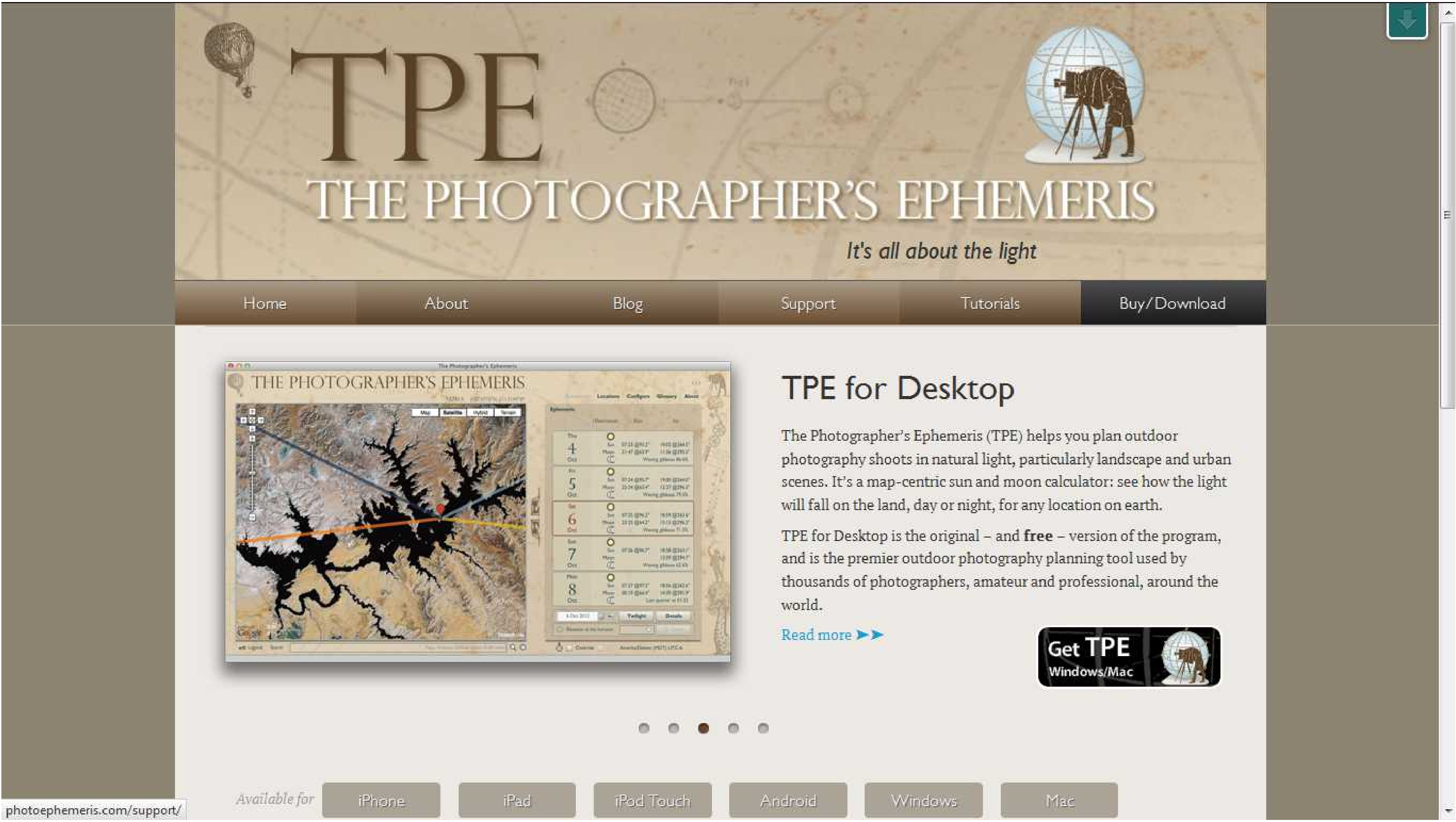
12
Today 16° 14°
Thu
Fri 19° 12°
Sat 17° 9°
Sun 17° 11°

Useful Apps


TPE
THE PHOTOGRAPHER'S EPHEMERIS
It's all about the light
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THE PHOTOGRAPHER'S EPHEMERIS
TPE for Desktop
The Photographer's Ephemeris (TPE) helps you plan outdoor photography shoots in natural light, particularly landscape and urban scenes. It's a map-centric sun and moon calculator: see how the light will fall on the land, day or night, for any location on earth.
TPE for Desktop is the original – and **free** – version of the program, and is the premier outdoor photography planning tool used by thousands of photographers, amateur and professional, around the world.
Read more
Get TPE
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Available for
iPhone
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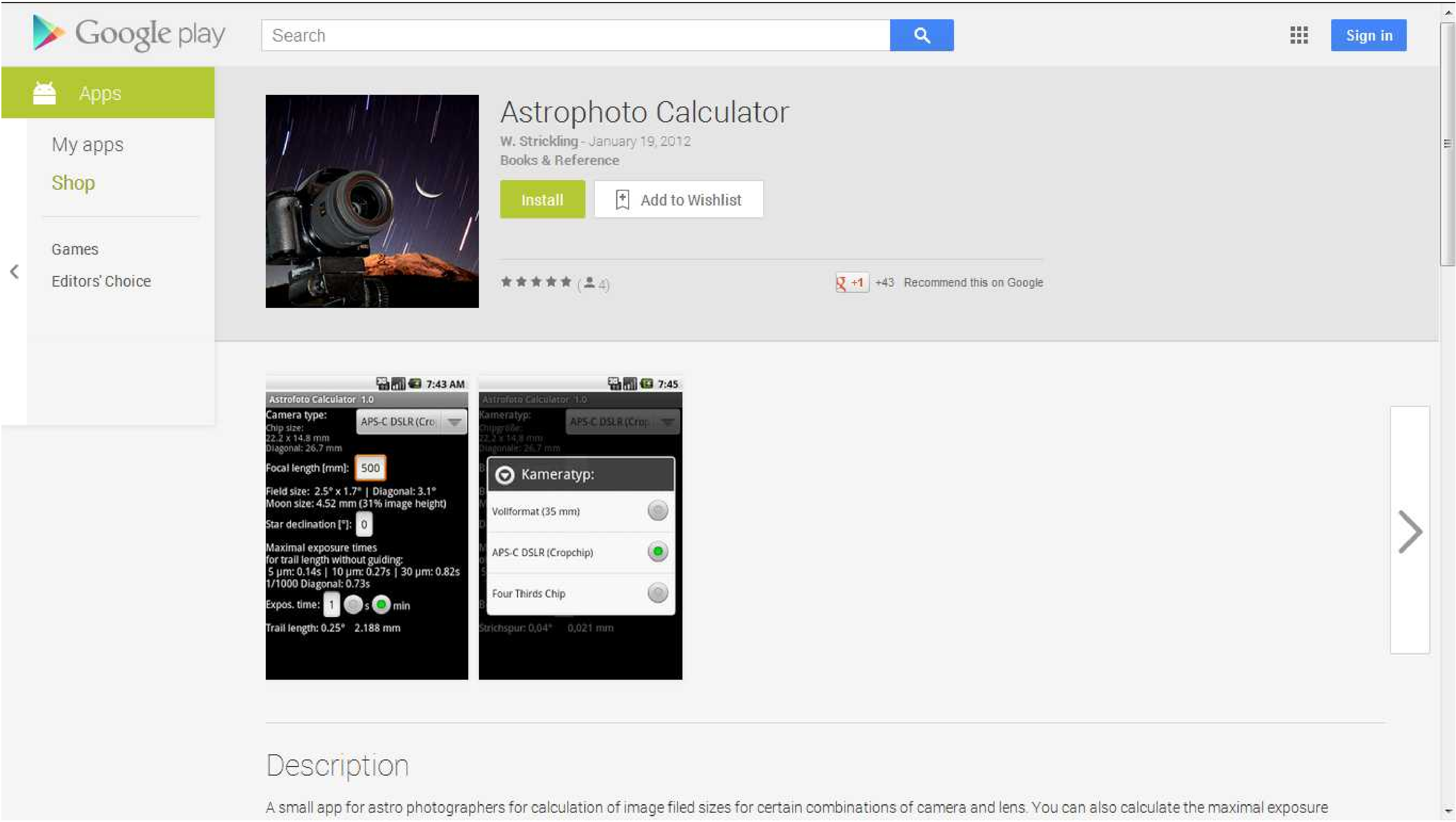
iPod Touch
Android
Windows
Mac
photoephemeris.com/support/

Useful Apps


Google play
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Sign in
Apps
My apps
Shop
Games
Editors' Choice
Astrophoto Calculator
W. Strickling - January 19, 2012
Books & Reference
Install
Add to Wishlist
+1 +43 Recommend this on Google
7:43 AM
Astrofoto Calculator 1.0
Camera type:
APS-C DSLR (Cro
Chip size:
22.2 x 14.8 mm
Diagonal: 26.7 mm
Focal length [mm]: 500
Field size: 2.5° x 1.7° | Diagonal: 3.1°
Moon size: 4.52 mm (31% image height)
Star declination [°]: 0
Maximal exposure times
for trail length without guiding:
5 µm: 0.14s | 10 µm: 0.27s | 30 µm: 0.82s
1/1000 Diagonal: 0.73s
Expos. time: 1 s min
Trail length: 0.25° 2.188 mm
7:45
Kameratyp:
Vollformat (35 mm)
APS-C DSLR (Cropchip)
Four Thirds Chip
Strichspur: 0,04° 0,021 mm
Description
A small app for astro photographers for calculation of image filed sizes for certain combinations of camera and lens. You can also calculate the maximal exposure

Useful Apps


ISS DETECTOR
SEE THE ISS WITH THE NAKED EYE
Know when and where to watch for the International Space Station and Iridium Flares
Free, easy to use, includes weather conditions, timings and a compass
Support for phones and tablets
See a demo on Youtube

Useful Apps

Google Mobile

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Google Sky Map

Explore the skies on your Android phone. ▷ Watch a video

Google Sky Map has been open sourced and released to the community as "Sky Map". Please go to the Google Play to download the latest version. For further information, please see the blog post.

Supported phones

All Android phones running Android 1.6 or higher.

+ FAQ

+ Troubleshooting

United States